

308-310页2-7,10,12.

310页8,9,14; 317页3,4,7,9,10.

317-318页1,5,11,13,14; 324页3.

308-2

Let $V = \mathbb{C}^{n \times n}$ with inner product $\langle A, B \rangle = \text{tr}(AB^*)$. For each $M \in V$, let $T_M \in \mathcal{L}(V) : A \mapsto MA$. Show that T_M is unitary iff M is unitary.

Proof: T_M is unitary iff $\langle T_M A, T_M B \rangle = \langle A, B \rangle$ for any A, B , i.e., $\text{tr}((MA)(MB)^*) = \text{tr}(AB^*)$, iff $\text{tr}(MCM^*) = \text{tr}(C)$ for any C . Since $\text{tr}(MCM^*) = \text{tr}(CMM^*)$, this is equivalent to $MM^* = I$ or M is unitary.

308-4

Let $U_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$.

(a) What is $U_\theta U_\phi$? (b) Show that $U_\theta^* = U_{-\theta}$; (c) Let ϕ be fixed, and $\mathcal{B} = \{\alpha_1, \alpha_2\}$ the orthonormal basis obtained by rotating $\{e_1, e_2\}$ through ϕ . If θ is another real number, what is the matrix of U_θ in $\mathcal{B}$?

Solution: (a) $U_{\theta+\phi}$. (b) By definition $U_\theta^* = \overline{U_\theta^T} = U_{-\theta}$.

(c) Clearly it is $U_{\theta-\phi}$.

308-5

Let $V = \mathbb{R}^3$, $W = \text{Span}\{\alpha, \beta\}$ where $\alpha = (1, 1, 1)$, $\beta = (1, 1, -2)$. Let $U \in \mathcal{L}(V)$ be the rotation through the angle θ , about the straight line through the origin, that is orthogonal to W . (choose any one orientation). Find the matrix of U in the standard basis.

Solution:

$$A = \begin{pmatrix} \frac{\cos \theta + 1}{2} & \frac{\cos \theta - 1}{2} & \frac{\sin \theta}{\sqrt{2}} \\ \frac{\cos \theta - 1}{2} & \frac{\cos \theta + 1}{2} & \frac{\sin \theta}{\sqrt{2}} \\ -\frac{\sin \theta}{\sqrt{2}} & -\frac{\sin \theta}{\sqrt{2}} & \cos \theta \end{pmatrix}$$

308-6

Let V be finite dimensional and W a subspace, then $V = W \oplus W^\perp$. Define a linear operator $U : \alpha \mapsto \beta - \gamma$ where $\alpha = \beta + \gamma \in W + W^\perp$.

(a) Prove that U is self-adjoint and unitary.

(b) If $V = \mathbb{R}^3$ and $W = \text{Span}\{(1, 0, 1)\}$, find the matrix of U .

Proof: (a) For $\alpha = \beta + \gamma$ and $\alpha' = \beta' + \gamma'$,

$$\langle U\alpha, \beta \rangle = \langle \beta - \gamma, \beta' + \gamma' \rangle = \langle \beta, \beta' \rangle - \langle \gamma, \gamma' \rangle = \langle \alpha, U\alpha' \rangle.$$

hence U is self-adjoint, and clearly $\|U\alpha\|^2 = \|\beta\|^2 + \|\gamma\|^2 = \|\alpha\|^2$ so U is unitary.

(b)

$$U = \begin{pmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

310-10

For each $\alpha, \beta \in V$, let $T_{\alpha, \beta} \in \mathcal{L}(V) : \gamma \mapsto \langle \gamma, \beta \rangle \alpha$. Show that

(a) $T_{\alpha, \beta}^* = T_{\beta, \alpha}$; (b) $\text{tr}(T_{\alpha, \beta}) = \langle \alpha, \beta \rangle$; (c) $T_{\alpha, \beta} T_{\gamma, \delta} = T_{\alpha, \langle \beta, \gamma \rangle \delta}$; (d) under what conditions is $T_{\alpha, \beta}$ self-adjoint?

Proof: (a) $\langle T_{\alpha, \beta} u, v \rangle = \langle \langle u, \beta \rangle \alpha, v \rangle = \langle u, \beta \rangle \langle \alpha, v \rangle = \langle u, T_{\beta, \alpha} v \rangle$.

(b) $\text{tr}(T_{\alpha, \beta}) = \sum \langle T_{\alpha, \beta} e_k, e_k \rangle = \sum \langle \alpha, e_k \rangle \langle e_k, \beta \rangle = \langle \alpha, \beta \rangle$.

(c) $T_{\alpha, \beta} T_{\gamma, \delta}(u) = T_{\alpha, \beta}(\langle u, \delta \rangle \gamma) = \langle \langle u, \delta \rangle \gamma, \beta \rangle \alpha = T_{\alpha, \langle \beta, \gamma \rangle \delta}(u)$.

(d) $T_{\alpha, \beta}$ is self-adjoint iff $T_{\alpha, \beta} = T_{\beta, \alpha}$, i.e., $\langle \gamma, \beta \rangle \alpha = \langle \gamma, \alpha \rangle \beta$ which is equivalent to α, β are linearly dependent.

310-12

In Ex6 we showed how to construct linear operators on V that are both self-adjoint and unitary, show that there are no others.

Proof: Note that if U is both self-adjoint and unitary, then $U^2 = U(U^*) = I$. Let $W = \text{Ker}(U - I)$ and $Z = \text{Ker}(U + I)$ then $(U - I)(U + I) = 0$ implies $Z \subset W^\perp$ and $W \subset Z^\perp$ so $W = Z^\perp$. Hence U is exactly the linear operator constructed from W .

310-8

If $\theta \in \mathbb{R}$, prove that the following are unitarily equivalent:

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}$$

Proof: Clearly the first matrix has eigenvalues $e^{i\theta}$ and $e^{-i\theta}$, and it is normal. Hence it is unitarily equivalent to the second one.

310-9

Let $T \in \mathcal{L}(V)$, $p_T(\alpha, \beta) = \langle T\alpha, \beta \rangle$. Let $U \in \mathcal{L}(V)$ and U^* is adjoint with respect to $\langle, \rangle$. Prove that U is unitary with respect to p_T iff $T = U^*TU$.

Proof: U is unitary iff

$$\langle T\alpha, \alpha \rangle = p_T(\alpha, \alpha) = p_T(U\alpha, U\alpha) = \langle TU\alpha, U\alpha \rangle$$

for any $\alpha \in V$, i.e. $\langle (T - U^*TU)\alpha, \alpha \rangle = 0$ for all $\alpha \in V$. Hence it is equivalent to $T = U^*TU$.

310-14

A rigid motion is a function $T : V \rightarrow V$ such that $\|T\alpha - T\beta\| = \|\alpha - \beta\| \forall \alpha, \beta \in V$.

(a) Let $V = \mathbb{R}^2$. Suppose T is a rigid motion such that $T(0) = 0$, show that T is linear and unitary.

(b) Use (a) to prove that every rigid motion of $\mathbb{R}^2$ is composed of a translation, followed by a unitary operator.

(c) Now show that a rigid motion of $\mathbb{R}^2$ is either a translation followed by a rotation, or a translation followed by a reflection and a rotation.

Proof: (a) Clearly T keeps the inner product. Expanding $\|T(\alpha x + \beta y) - \alpha T(x) - \beta T(y)\|^2$ gives $T(\alpha x + \beta y) = \alpha T(x) + \beta T(y)$ so T is linear. Therefore T is unitary.

(b) If T is a rigid motion, then $T - T(0)$ is also rigid, hence a unitary operator.

(c) Use (b) and the fact that unitary operators on $\mathbb{R}^2$ are either rotations or reflection followed by a rotation.

317-3

For $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{pmatrix}$ there is a real orthogonal matrix P such that $P^t A P = D$ is diagonal. Find such a diagonal matrix D .

Solution:

$$D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \frac{9+\sqrt{105}}{2} & 0 \\ 0 & 0 & \frac{9-\sqrt{105}}{2} \end{pmatrix}$$

317-4

Let $V = \mathbb{C}^2$ and $T \in \mathcal{L}(V)$ represented by $A = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$. Show that T is normal, and find an orthonormal basis for V , consisting of eigenvalues of T .

Proof: Clearly $AA^* = A^*A = 2I$ so T is normal. We have the orthogonal basis $e_1 = \frac{1}{\sqrt{2}}(1, -1)$ and $e_2 = \frac{1}{\sqrt{2}}(1, 1)$.

317-7

Let $T \in \mathcal{L}(V)$ be positive and unitary, show that $T = I$.

Proof: Since T is positive, T is diagonalizable and has only positive eigenvalues. Hence T is unitary implies the only eigenvalue of T is 1, so $T = I$.

317-9

Prove that a real symmetric matrix has a real symmetric cube root.

Proof: A real symmetric matrix can be unitarily diagonalized, and a diagonal matrix has a cube root.

317-10

Prove that every positive matrix is a square of a positive matrix.

Proof: Since it is self-adjoint, we can unitarily diagonalize it $A = PDP^*$, such that $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ and $\lambda_i \geq 0$. Then we have $D^{1/2} = \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n})$, such that $A = (PD^{1/2}P^*)^2$, and $PD^{1/2}P^*$ is positive.

317-1

For each A find a real orthogonal P such that P^tAP is diagonal;

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}$$

Solution:

$$P = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}.$$

317-5

Give an example of a 2×2 matrix A such that A^2 is normal but A is not normal.

Solution: Consider $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$.

318-11

Prove that a normal and nilpotent operator is zero.

Proof: If N is normal and nilpotent, we show that $\text{Ker} N^2 = \text{Ker} N$: If $N^2x = 0$ then $\|N(Nx)\| = 0$ implies $\|N^*(Nx)\| = 0$ so $\langle Nx, Nx \rangle = \langle x, N^*Nx \rangle = 0$. Hence $Nx = 0$.

Therefore $\text{Ker} N^k = \text{Ker} N$ for normal N . Since N is nilpotent, $\text{Ker} N = V$ so $N = 0$.

318-13

Let T be a normal operator on a finite dim complex space. Prove that there is a polynomial $f \in \mathbb{C}[x]$ such that $T^* = f(T)$.

Proof: Since T commutes with T^* , they can be simultaneously unitarily diagonalized, hence by Lagrange interpolation such a polynomial exists.

318-14

If two normal operators commute, prove that their product is normal.

Proof: We can assume the field is $\mathbb{C}$, then two normal operators commute imply they can be simultaneously unitarily diagonalized, so their product is still normal. For real operators we can view their matrix as complex matrices.

324-3

Let $A = \begin{pmatrix} 1 & i \\ -i & 2 \end{pmatrix}$ and let g be the form $g(X, Y) = Y^*AX$. Is g an inner product?

Solution: For $X = (u, v) \in \mathbb{C}^{2 \times 1}$, $g(X, X) = |u|^2 + 2|v|^2 \geq 0$ so g is an inner product.